

JavaScript Day 1 - Assignment

Name: _____

Date: _____

Instructions

- Complete all the problems below
 - Write your code in a JavaScript file or directly in the browser console
 - Test your solutions to verify they work correctly
 - Show all your work and include comments explaining your logic
 - Submit your completed assignment by the deadline
-

Part A: Variable Declaration and Data Types (20 points)

Problem 1: Create Variables (4 points)

Declare variables with appropriate names and assign the following values:

- Your first name (string)
- Your age (number)
- Whether you're a student (boolean)
- Your favorite color (string)
- Number of siblings you have (number)

```
javascript
```

```
// Write your code here
```

Problem 2: Type Checking (4 points)

For each variable you created in Problem 1, use the `typeof` operator to check and log its data type.

```
javascript
```

```
// Write your code here
```

Problem 3: Understanding Undefined and Null (4 points)

a) Declare a variable named `score` without assigning a value. Log its value and type. b) Create a variable named `winner` and assign it the value `null`. Log its value and type. c) Explain in your own words: What's the difference between undefined and null?

```
javascript
```

```
// Write your code here
```

Explanation:

[Write your explanation here]

Problem 4: Working with Strings (4 points)

Create three variables:

- `firstName` with your first name
- `lastName` with your last name
- `fullName` that combines both with a space in between

Log the `fullName` to the console.

```
javascript
```

```
// Write your code here
```

Problem 5: Type Conversion Challenge (4 points)

What will be the output of the following? Try to predict before running the code:

```
javascript
```

```
let a = "5";  
let b = 3;  
let result = a + b;  
console.log(result);  
console.log(typeof result);
```

Your prediction:

[Write your prediction here]

Actual output:

[Write the actual output after running the code]

Explanation: Why did this happen?

[Write your explanation here]

Part B: Arithmetic Operations (25 points)

Problem 6: Basic Calculator (5 points)

Create two variables `num1 = 45` and `num2 = 12`. Calculate and store:

- Sum
- Difference
- Product
- Quotient
- Remainder (using modulo)

Log all five results with descriptive labels.

```
javascript
```

```
// Write your code here
```

Problem 7: Temperature Converter (5 points)

Write code to convert temperature from Celsius to Fahrenheit.

- Formula: `Fahrenheit = (Celsius × 9/5) + 32`
- Test with `Celsius = 25`
- Log the result: "25°C is equal to __°F"

```
javascript
```

```
// Write your code here
```

Problem 8: Circle Calculations (5 points)

Given a circle with radius = 7:

- Calculate the circumference: $2 \times \pi \times \text{radius}$ (use `Math.PI` for π)
- Calculate the area: $\pi \times \text{radius}^2$
- Log both results rounded to 2 decimal places (Hint: use `.toFixed(2)`)

```
javascript
```

```
// Write your code here
```

Problem 9: Shopping Bill Calculator (5 points)

You bought:

- 3 notebooks at ₹45 each
- 2 pens at ₹15 each
- 1 backpack at ₹850

Calculate:

- a) Subtotal (total before discount)
- b) Discount amount (10% off on subtotal)
- c) Final total (after discount)

```
javascript
```

```
// Write your code here
```

Problem 10: Even or Odd Checker (5 points)

Create a variable `number = 17`. Use the modulo operator (%) to determine if it's even or odd.

- If `number % 2 === 0`, it's even
- Otherwise, it's odd

Store the result in a variable and log it.

```
javascript
```

```
// Write your code here
```

Part C: Operator Precedence (15 points)

Problem 11: Predict the Output (5 points)

For each expression, predict the result before running the code:

```
javascript
```

```
// a)
```

```
let result1 = 10 + 5 * 2;
```

```
// Your prediction: _____
```

```
// Actual output: _____
```

```
// b)
```

```
let result2 = (10 + 5) * 2;
```

```
// Your prediction: _____
```

```
// Actual output: _____
```

```
// c)
```

```
let result3 = 20 / 4 + 3 * 2;
```

```
// Your prediction: _____
```

```
// Actual output: _____
```

```
// d)
```

```
let result4 = 2 ** 3 + 5;
```

```
// Your prediction: _____
```

```
// Actual output: _____
```

```
// e)
```

```
let result5 = (8 + 2) / 5 * 3;
```

```
// Your prediction: _____
```

```
// Actual output: _____
```

Problem 12: Fix the Expression (5 points)

The following expression is supposed to calculate: "Add 10 and 20, then multiply by 5, then divide by 3"

Current code:

```
javascript
```

```
let result = 10 + 20 * 5 / 3;
```

This gives the wrong answer due to operator precedence. Fix it using parentheses:

```
javascript
```

// Write your corrected code here

Problem 13: Create a Formula (5 points)

Write an expression to calculate the following using operator precedence:

- Take the number 15
- Add 5 to it
- Multiply the result by 3
- Subtract 10 from that
- Divide by 4

Do this in ONE line using proper parentheses. Store in a variable called `finalResult`.

javascript

// Write your code here

Part D: Assignment Operators (15 points)

Problem 14: Bank Account Simulation (8 points)

You have a bank account with ₹5000.

- You receive your salary: add ₹15000 (use `+=`)
- You pay rent: subtract ₹8000 (use `-=`)
- You get a bonus: add ₹5000 (use `+=`)
- You spend on shopping: subtract ₹3000 (use `-=`)

Calculate the final balance and log it after each transaction.

javascript

// Write your code here

Problem 15: Compound Interest (7 points)

You invest ₹10000 at 10% annual interest, compounded yearly for 3 years.

- Year 1: amount `*= 1.10`

- Year 2: amount *= 1.10
- Year 3: amount *= 1.10

Use the `*=` operator for each year and log the amount after each year.

```
javascript
```

```
// Write your code here
```

Part E: Increment and Decrement Operators (15 points)

Problem 16: Pre vs Post Increment (6 points)

Predict the output of the following code, then run it to verify:

```
javascript
```

```
let x = 10;
let y = x++;
console.log("x:", x, "y:", y);

let a = 10;
let b = ++a;
console.log("a:", a, "b:", b);
```

Your predictions:

First console.log - x: ____, y: ____

Second console.log - a: ____, b: ____

Actual output:

[Write the actual output here]

Explanation: Why are the results different?

[Write your explanation here]

Problem 17: Countdown Timer (5 points)

Create a variable `countdown = 10`. Using the decrement operator, simulate a countdown from 10 to 1. Log each

number.

(Hint: You'll learn loops later, but for now, just decrement and log 10 times manually)

```
javascript
```

```
// Write your code here
```

Problem 18: The Tricky Problem (4 points)

Solve the problem from class:

```
javascript
```

```
let num = 5;
```

```
let newNum = num++; // What are num and newNum now?
```

```
newNum = ++num;    // What are num and newNum now?
```

Show the values of both variables after each line:

After line 2: num = ____, newNum = ____

After line 3: num = ____, newNum = ____

Part F: Working with NaN (10 points)

Problem 19: Understanding NaN (5 points)

What will be the result of each operation? Write "NaN" or the numeric result:

```
javascript
```



```
// a)
let result1 = 0 / 0;
// Result: _____

// b)
let result2 = "hello" * 5;
// Result: _____

// c)
let result3 = NaN + 10;
// Result: _____

// d)
let result4 = Math.sqrt(-1);
// Result: _____

// e)
console.log(typeof NaN);
// Result: _____
```

Problem 20: Detecting NaN (5 points)

Write code to check if the following values are NaN using `Number.isNaN()`:

```
javascript

let value1 = 100;
let value2 = "test" / 2;
let value3 = 0 / 0;
let value4 = "123";

// Write your code to check each value here
```

Part G: Real-World Applications (Bonus: 10 points)

Problem 21: Grade Calculator (5 points)

Calculate your average grade from 5 subjects:

- Math: 85
- Science: 92
- English: 78

- History: 88
- Computer: 95

Store each subject score in a variable, calculate the average, and log it with a message: "Your average grade is: ___"

```
javascript
```

```
// Write your code here
```

Problem 22: Age Calculator (5 points)

Write code that:

- Stores the current year (2024) in a variable
- Stores your birth year in a variable
- Calculates your age (or age you'll turn this year)
- Logs: "In [current year], you are [age] years old"

```
javascript
```

```
// Write your code here
```

Part H: Challenge Problems (Bonus: 10 points)

Problem 23: Swap Values Without a Third Variable (5 points)

Given two variables:

```
javascript
```

```
let a = 10;
```

```
let b = 20;
```

Swap their values WITHOUT using a third variable. (Hint: use arithmetic operators)

After swapping: `a = 20` and `b = 10`

```
javascript
```

```
// Write your code here
```

Problem 24: Digit Sum Calculator (5 points)

Given a three-digit number like 456, calculate the sum of its digits ($4 + 5 + 6 = 15$).

Hint:

- Use `/ 100` and `Math.floor()` to get the first digit
- Use `% 100` to remove the first digit
- Use `/ 10` and `Math.floor()` to get the second digit
- Use `% 10` to get the last digit

```
javascript

let number = 456;
// Write your code here
```

Submission Checklist

Before submitting, make sure:

- ☐ All problems are attempted
- ☐ Code is properly commented
- ☐ All predictions and explanations are filled in
- ☐ Code has been tested in the browser console
- ☐ Output matches expected results
- ☐ Your name and date are filled in at the top

Grading Rubric

Section	Points	Score
Part A: Variables & Data Types	20	
Part B: Arithmetic Operations	25	
Part C: Operator Precedence	15	
Part D: Assignment Operators	15	
Part E: Increment/Decrement	15	

Section	Points	Score
Part F: Working with NaN	10	
Part G: Real-World Applications (Bonus)	10	
Part H: Challenge Problems (Bonus)	10	
Total	100 + 20 bonus	

Tips for Success

1. **Test as you go:** Don't wait until the end to test your code
2. **Use `console.log()`:** Print intermediate results to understand what's happening
3. **Read error messages:** They often tell you exactly what's wrong
4. **Experiment:** Try different values to see how your code behaves
5. **Ask for help:** If you're stuck for more than 15 minutes, reach out!

Additional Resources

- [MDN Web Docs - JavaScript Basics](#)
- [JavaScript.info - Variables](#)
- [W3Schools - JavaScript Operators](#)

Good luck and happy coding! 🚀

Remember: The goal is to learn, not just to finish. Take your time, understand each concept, and don't hesitate to ask questions!